Action on Categories with Adjoints

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1 Ideas

I'm starting to grow convinced that the action will come from applying the "classifying space" functor to a map of posets. More specifically, there's many colimit diagrams describing $c_k^{n-\mathsf{adj}}$ and they are arranged in a poset. I think roughly moving through that poset means describing the same cell by different formulae. This should correspond to the automorphisms of categories with adjoints.

If my "signed Schubert cells" are correct, then $BO(n)$ is the realization of a poset. A reasonnable question is: can I find a similar poset for $PL(n)$? There are realizations by posets for $BPL(n)$ (triangulation/ball decompositions of the marked sphere) but they seem too flexible. Maybe I need to reduce to only some triangulations that are related to the signed Schubert cell picture. No idea how these pictures are related though.

2 Signed Schubert Cells

Fix $n \geq 0$. We write $\mathbf{n} = \{0, \ldots, n-1\}$.

Definition 2.1 (Shape and Submatrices)**.** A shape is an strictly increasing sequence of integers $0 \leq \sigma_1 < \ldots < \sigma_n$. A submatrix of this shape is the data of subsets $I \subseteq \mathbf{n}$ and $J \subseteq \mathbb{N}$ such that $|I| = |J|$ and $\max(J) < \sigma_{\min(I)}$. We denote by $\mathsf{Sub}(\sigma)$ the set of submatrices of shape σ.

Definition 2.2 (Cell)**.** Let σ be a shape and $\epsilon \colon \mathsf{Sub}(\sigma) \to \{-, 0, +\}$ be a function. The signed Schubert cell $e_{\sigma,\epsilon}$ is the subset of Gr_n of all subspaces which row-reduce to σ in the canonical basis and for which the sign of the determinant of the submatrices are given by ϵ.

Lemma 2.1

All signed Schubert cells $e_{\sigma,\epsilon}$ are either empty or homeomorphic to $\mathbb{R}^m$ for some m.

Proof: Idea: filration on each coefficient. Each time the fiber should be either: empty, a point or an open interval. This is essentially because we have a bunch of **linear** inequalities on the coefficients of the matrix, which we can rewrite as $\min < x < \max$, or equalities that have unique solutions.

Take all submatrices involving some fix x_{ij}, *i.e.* with $(i, j) \in I \times J$, with a non-zero minor at this entry. We have either of the following cases:

- one of them is sent to 0 by the sign function ϵ, in which case x_{ij} is fixed to a single value, or no value.
- otherwise, each of them gives an open inequality on x_{ij}, the collection of which results in an open interval of possibilities, potentially an empty one.

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This effectively proves that the signed Schubert cells are "elementary sets" in the sense of Aaron Peterson's 291 course, with any ordering on the coordinates.

Lemma 2.2 (Faces of Schubert Cells)

When x_{ij} approaches its upper limit, it will approach a collection of inequalities. These effectively become equalities, the corresponding submatrices go from $\pm$ to 0. When "1 approaches 0", we have a reduction operation also giving us a single face (that's why we need all the determinants). Hence a "face" is choosing a subset of the $\pm$ submatrices and making them 0. There's potentially $2^{\#\text{ of }\pm}$ faces, but some of them may be empty.

What is the link? Suppose A is a face of B.

- Does all of A touches B? Yes, do some infenitesimal perturbation of the coordinates of A, and you'll land in B.
- Is the link conctractible? Yes I think so.

Then this means that the homotopy type of $BO(n)$ is the homotopy colimit of the poset of (non-empty) cells.

3 Action on Categories with Adjoints

- **Case** $n = 1$: We have $BO(1) = B(\mathbb{Z}/2\mathbb{Z})$ which acts by taking "op" on usual ∞-categories. Say a 1-cell is sent to "op" exactly when the unique sign is negative. Each "2-cell" is of the form $10\ldots0\pm0\ldots0\pm0\ldots0$. These have 3 faces. Say $-+$ gives a shortened $-$, a long $0+$ and a long -0. It's easy to see there's always an even number of $-$ signs, so the action is well-defined on the 2-cells. It's easy to see how it extends to all cells as these 2-cells are sent to identities.

- **Case** $n = 2$: